MAX PLANCK INSTITUTE
FOR BIOGEOCHEMISTRY

Lagged emissions from tree mortality dominate the inter-annual variations of the global CO_2 growth rate

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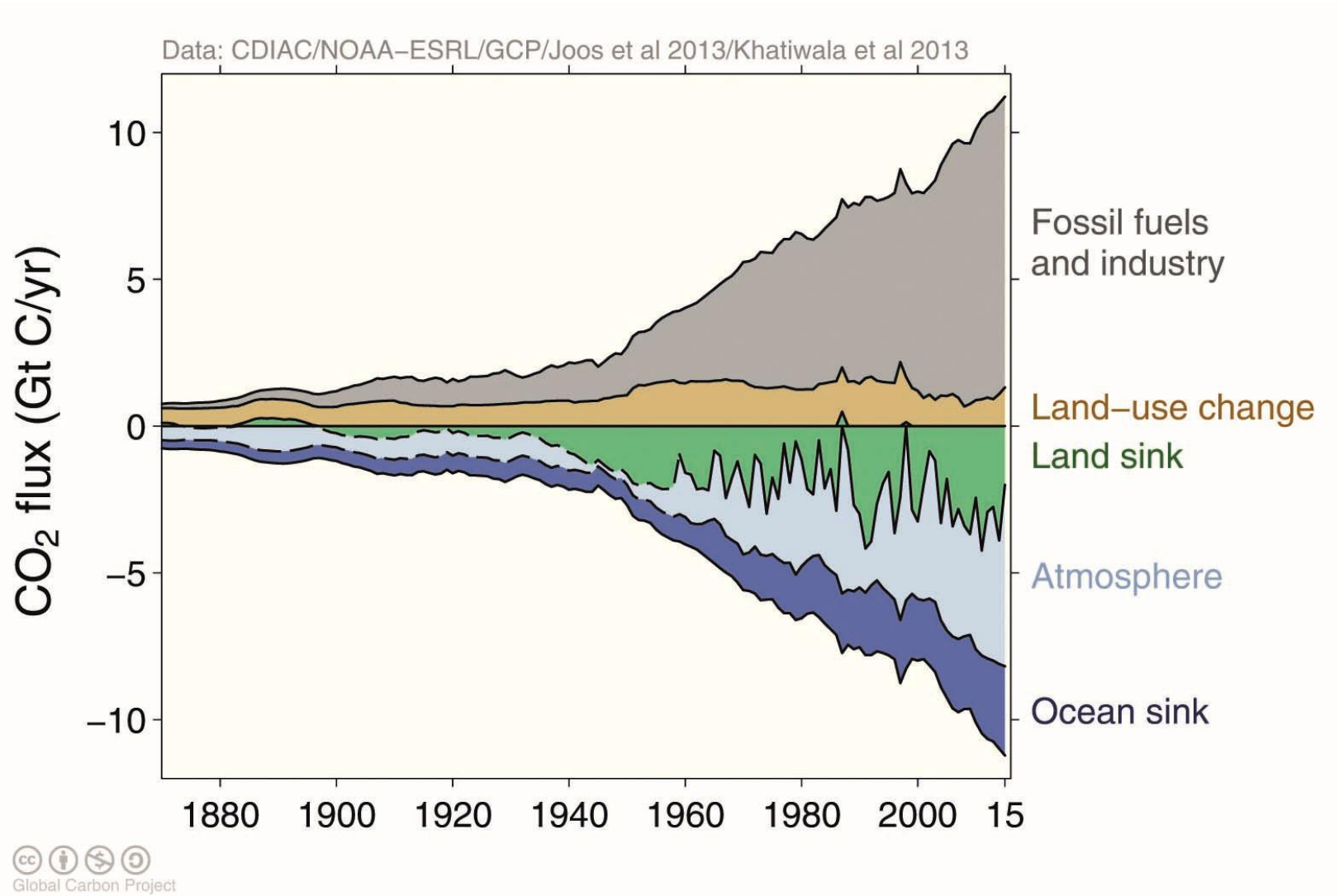
EEBIOMASS

Global Carbon Cycle over time



(The Annual Global Carbon Budget)

Scientific Questions

$$CGR = FF - S_{ocean} - S_{land}$$

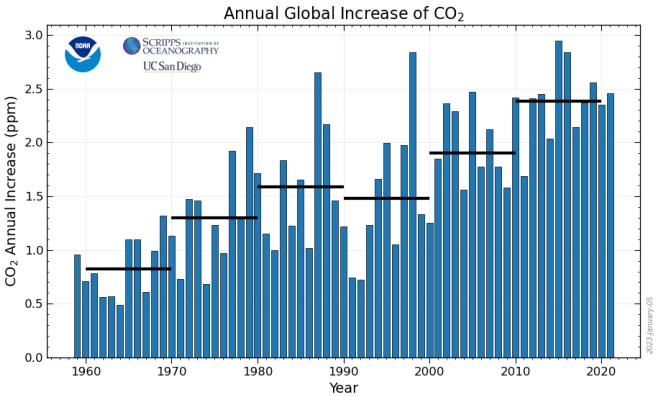
CO_2 growth rate in the atmosphere

Fossil Fuel emission

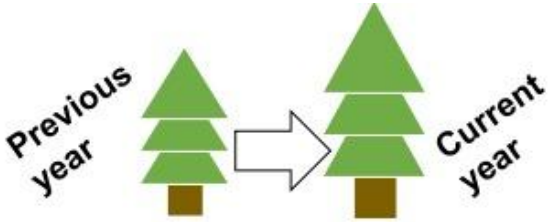
Net ocean sink

Net land sink

In situ CO_2 observations



Satellite-derived carbon stock changes



- Photosynthesis
- Respiration and Decay

- How does the plant biomass dynamics affect CGR?
- Have these plant contributions changed over time?

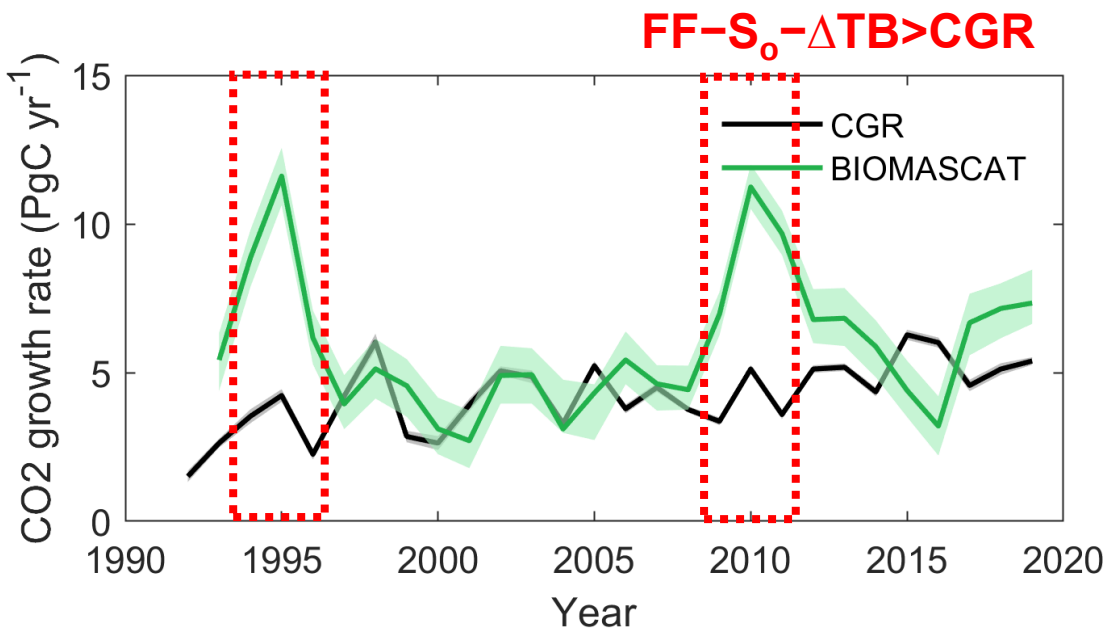
Direct comparison between CGR and biomass change

- **Assumption I:**

  All the live biomass loss cause instantaneous CO_2 emission to the atmosphere (‘direct’ or ‘immediate’ effect)

$$\mathbf{CGR \quad ? \quad FF - S_{ocean} - \Delta TB}$$

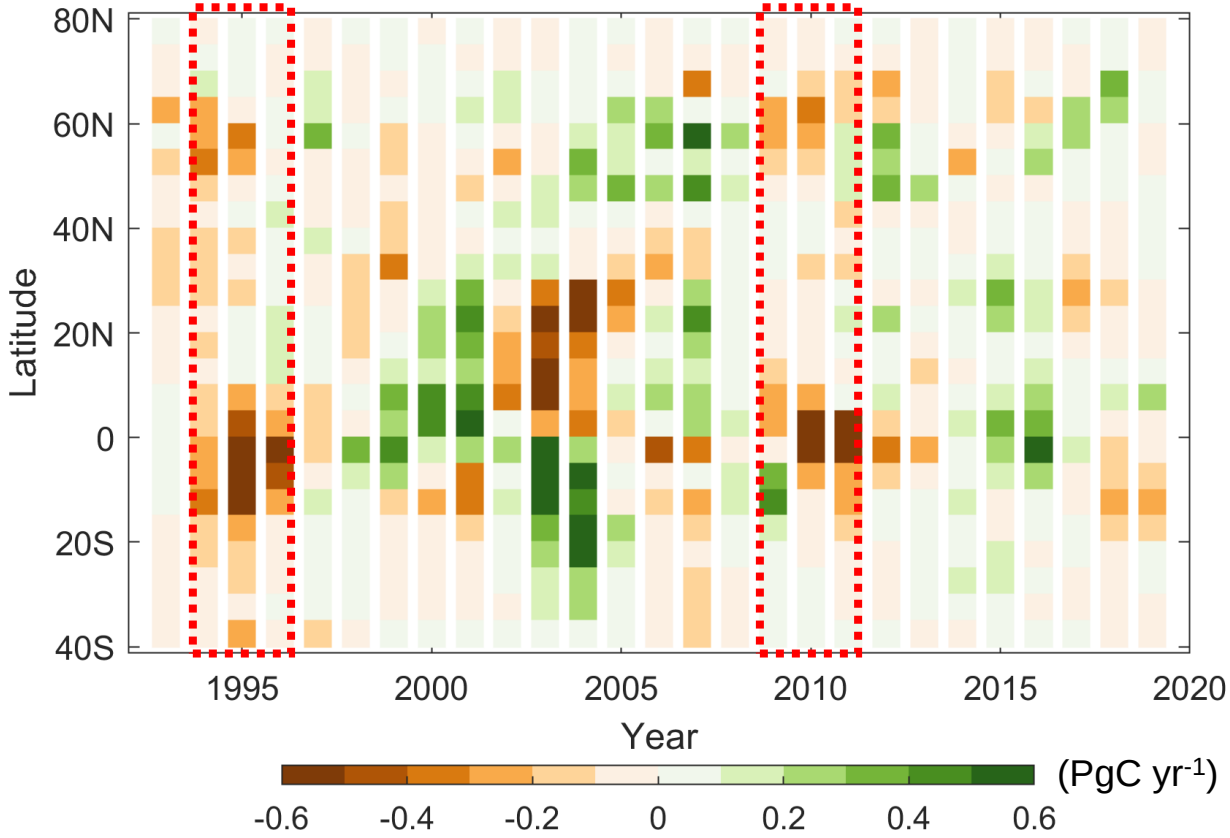
CGR	?	FF	S_{ocean}	ΔTB
NOAA Global mean		National fossil CO_2 emission	Global ocean biogeochemistry models	C-band scatterometer biomass product (Santoro et al., 2022)



**TB: Total Biomass = Above-ground biomass (AGB) + below-ground biomass (BGB)*

Satellite-derived carbon stock changes

We calculate the total biomass changes for each 10º latitudinal zones



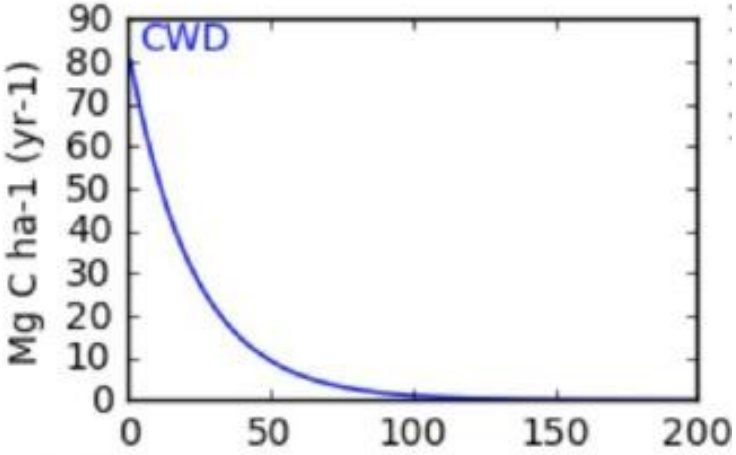
More than 4 PgC yr^{-1} biomass losses in the areas around 20ºS-10ºN latitudes in 1995/96 and 2010/11

Lagged decomposition of Coarse Woody Debris (CWD)

- Carbon within dead trees or branches, which forms CWD or enters into the soil does not lead to instantaneous CO_2 emission.

- It takes an amount of time for CWD to be decomposed by microbial processes.

CWD decomposition curve



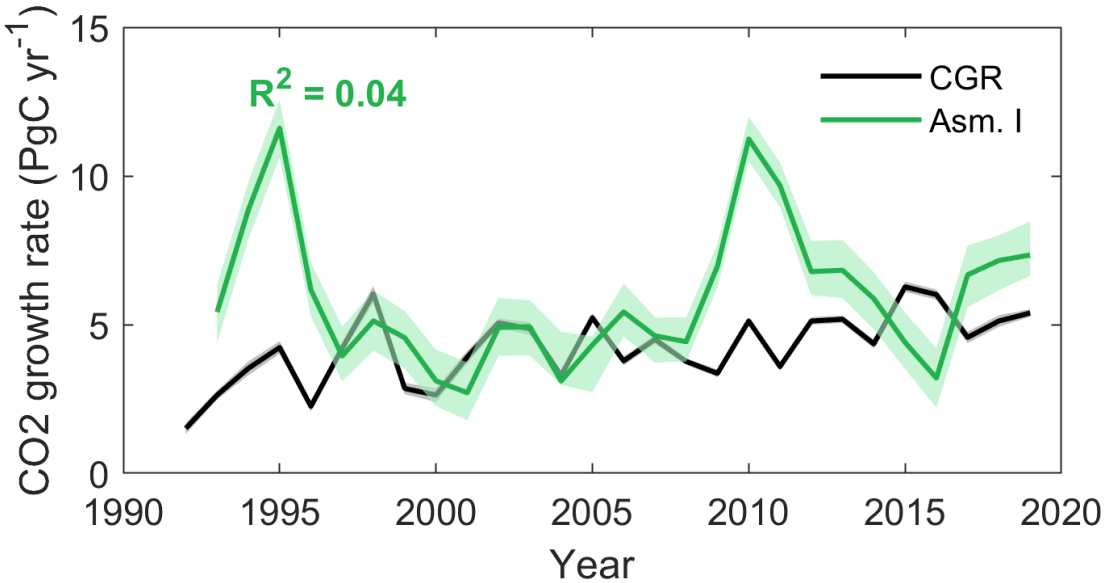
Lagged decomposition of Coarse Woody Debris (CWD)

- **Assmuption I:**

  All the live biomass loss cause instantaneous CO_2 emission to the atmosphere ('direct' or 'immediate' effect)

- **Assumption II:**

  All the live biomass loss become CWD and cause a lagged CO_2 emission from decaying CWD and SOC ('indirect' or 'lagged' effect)



Lagged decomposition of Coarse Woody Debris (CWD)

- **Assumption II:**

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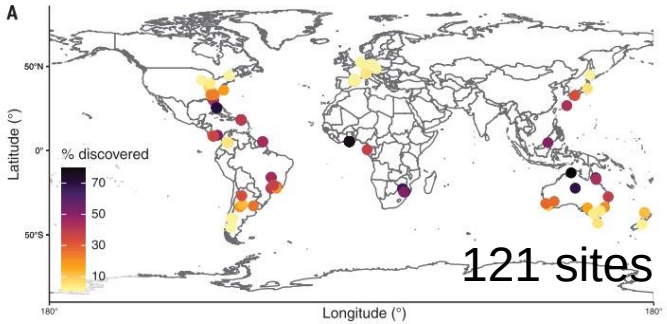
The change in carbon stocks of CWD and soil organic carbon (SOC) over time on a **yearly time step** at **each grid cell**

$$dCWD/dt = I(t) - (1-f)kCWD(t-1)$$

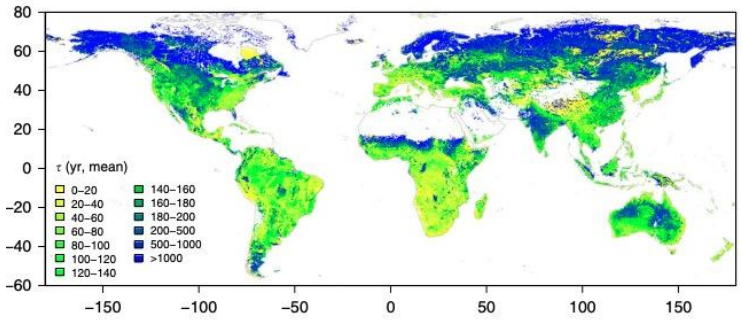
$$dSOC/dt = fkCWD(t) - SOC(t-1)/\tau_{SOC}$$

I(t) is the net live biomass change, *k* is CWD decomposition rate, τ_{SOC} is the SOC turnover time

CWD decomposition rate k
(Zanne et al., 2022)



SOC Turnover time
(Luo et al., 2019)



Lagged decomposition of Coarse Woody Debris (CWD)

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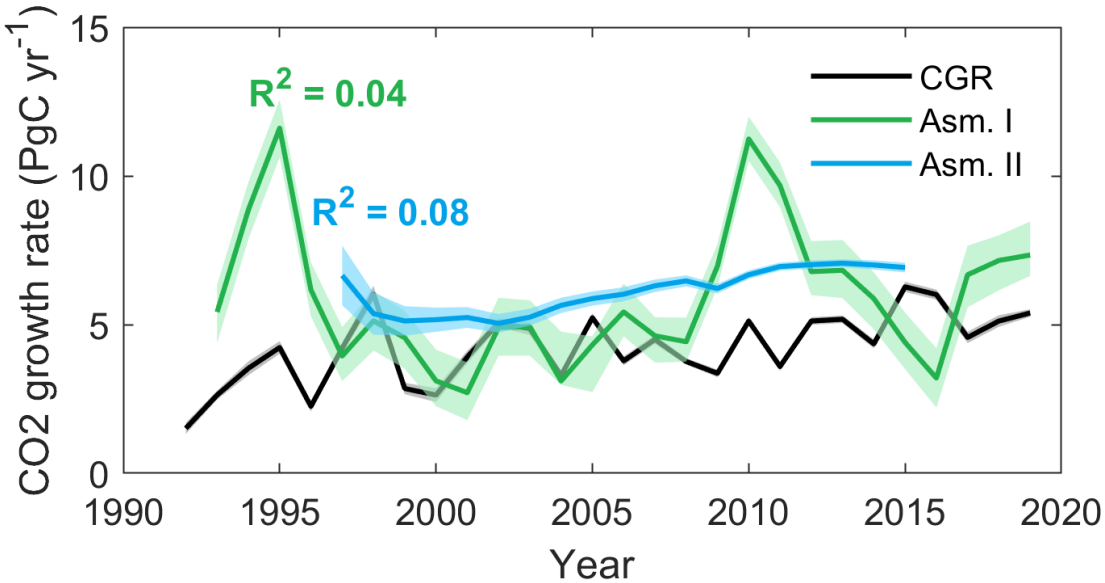
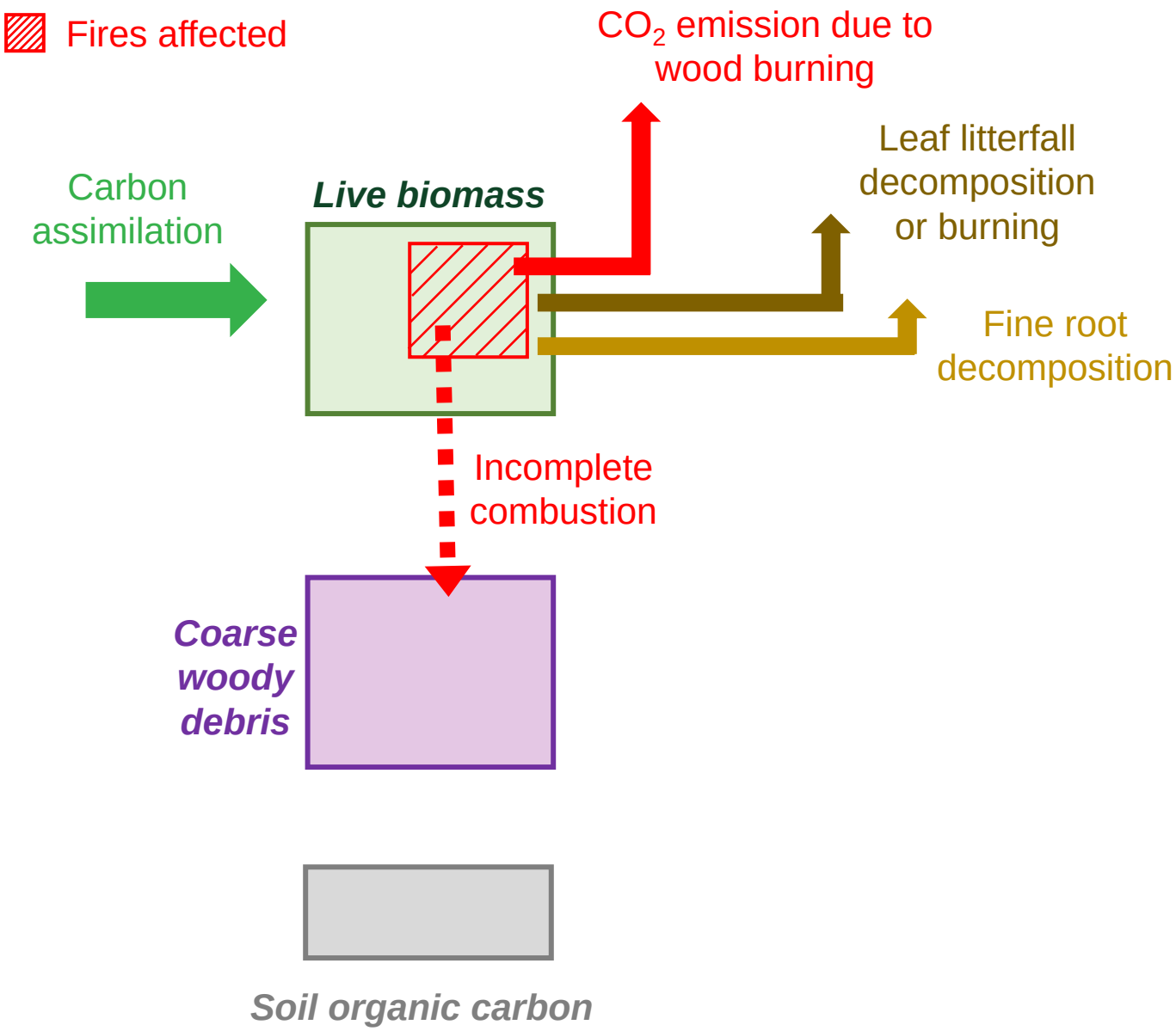


Data-driven three-pool box model



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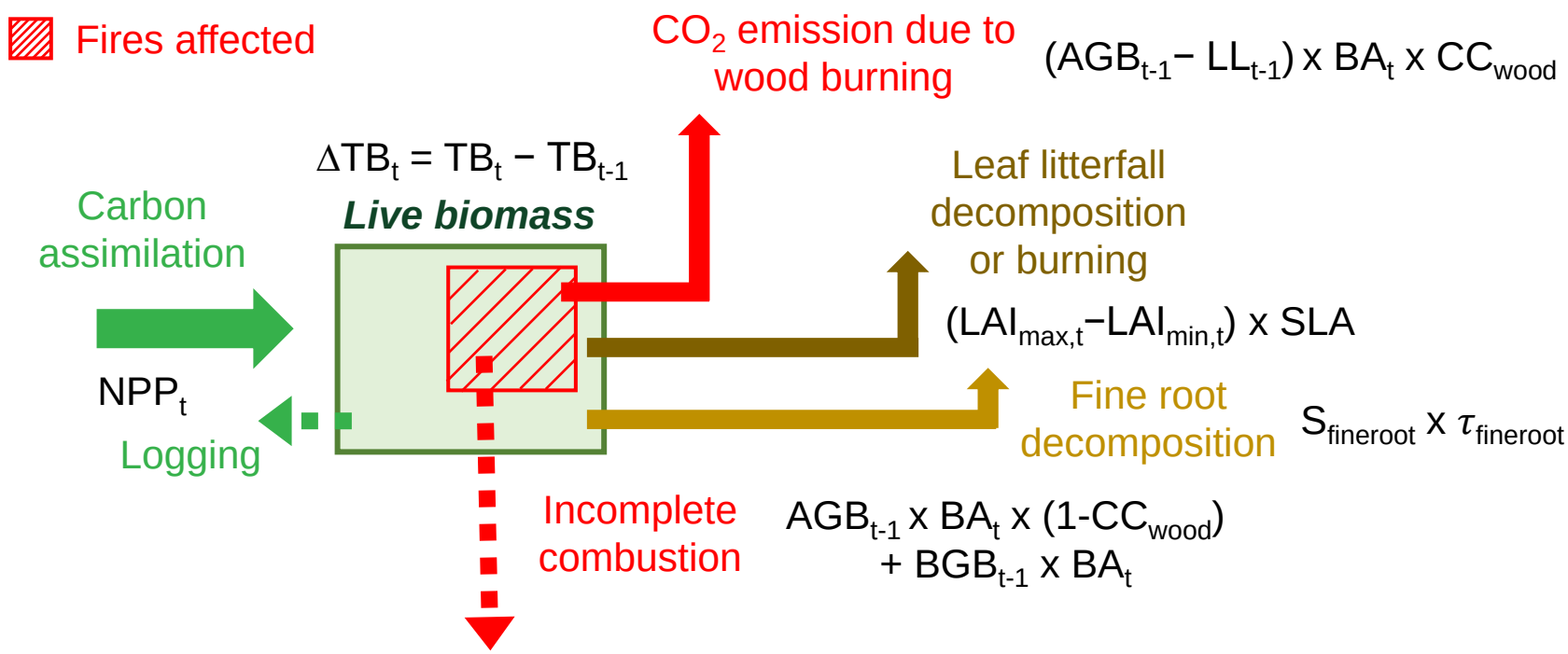


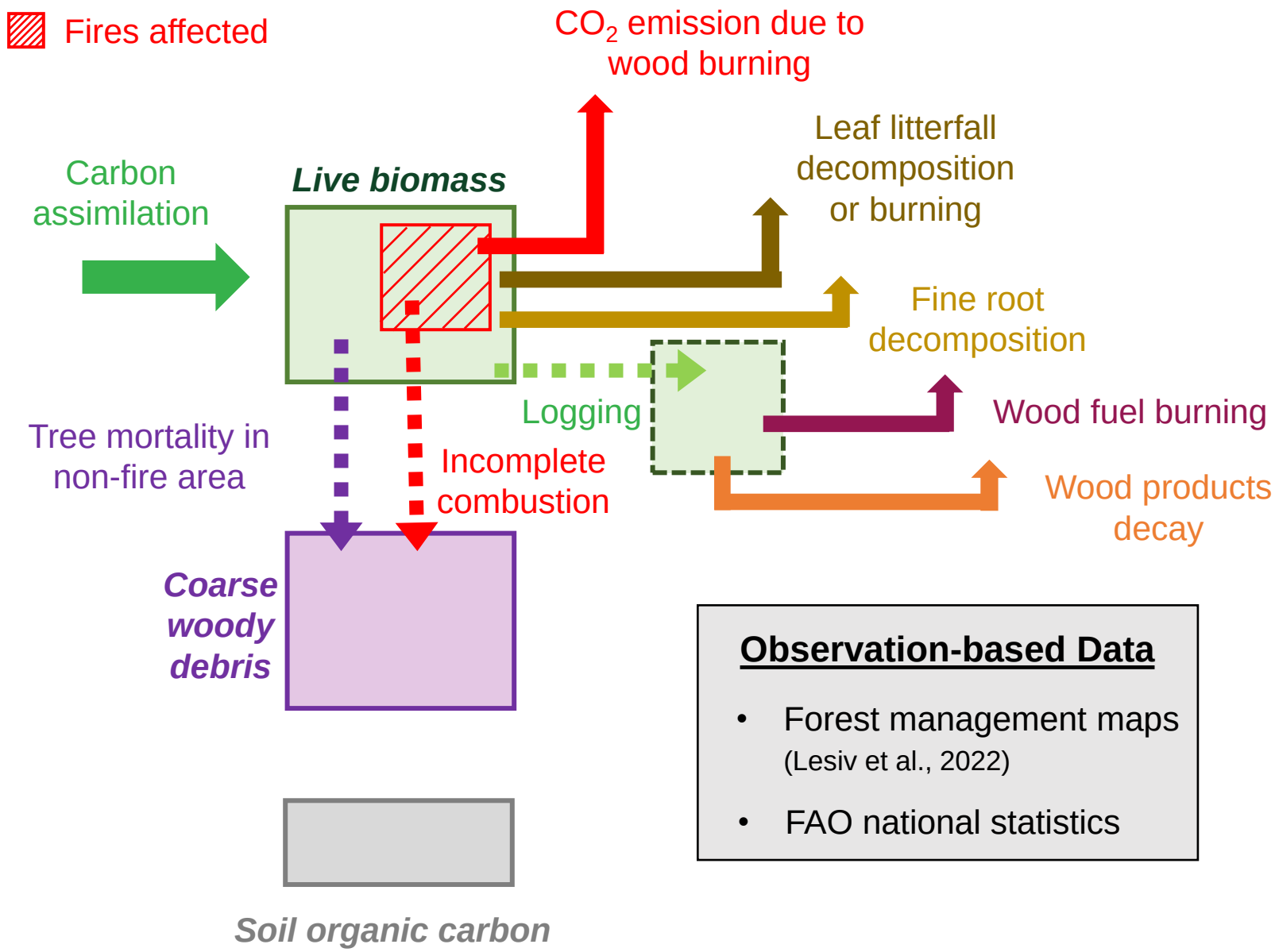
Satellite-based Data

- BIOMASCAT AGB
- GIMMS NPP
- GIMMS leaf area
- GFED4 burned area

Static Parameters

- AGB/BGB (Spawn et al., 2020)
- Specific Leaf Area (Butler et al., 2017)
- Combustion Completeness (CC) for wood (van Wees et al., 2022)
- Fine root allocation and turnover time (Bloom et al., 2016)

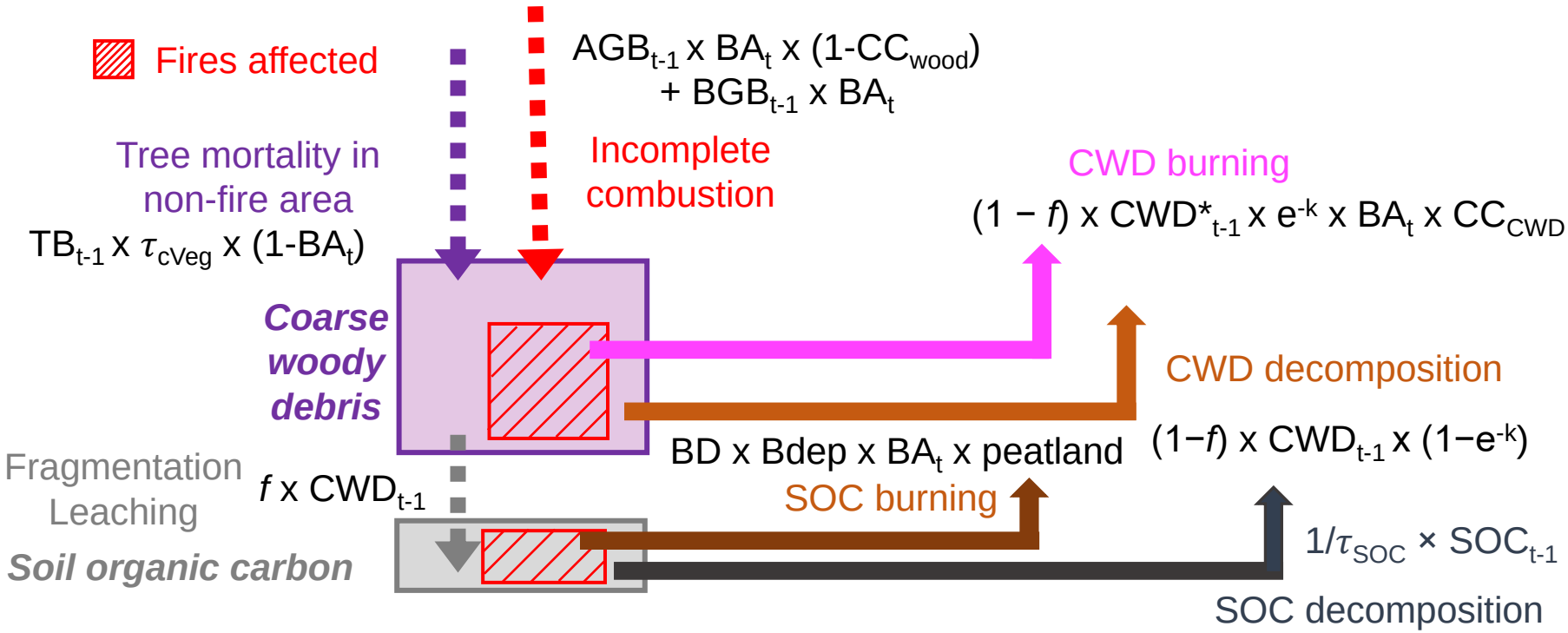
Data-driven three-pool box model



Observation-based Data

- Forest management maps (Lesiv et al., 2022)
- FAO national statistics

Data-driven three-pool box model



Satellite-based Data

- GFED4 burned area
- Burned depth

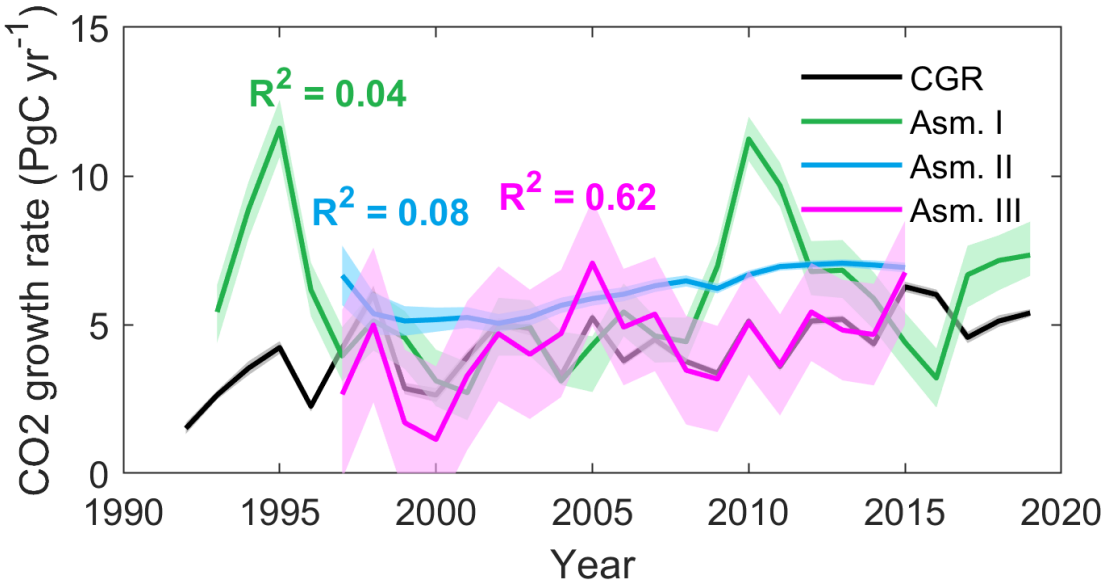
Static Parameters

- Combustion Completeness (CC) for CWD (van Wees et al., 2022)
- CWD decomposition rate (Zanne et al., 2022)
- SOC turnover rate (Luo et al., 2019)

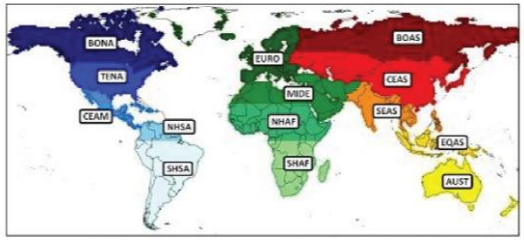
Data-driven three-pool box model

- **Assumption I:** ('direct' or 'immediate' effect)
- **Assumption II:** ('indirect' or 'lagged' effect)
- **Assumption III:** (Both 'immediate' & 'lagged' effect)

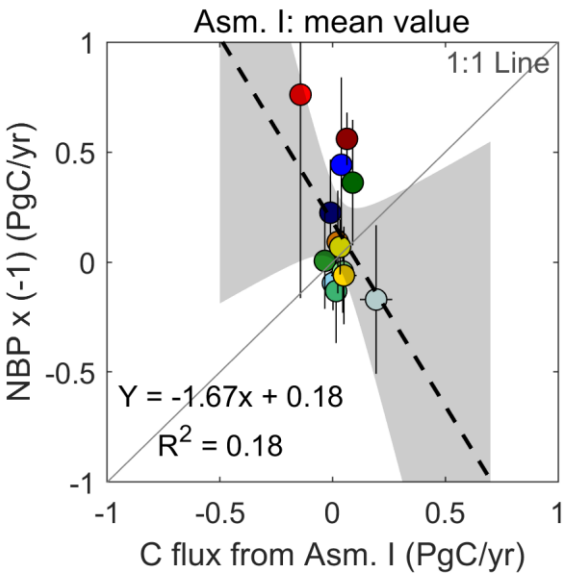
  Part of live biomass loss by fires cause instantaneous CO_2 emission, the remaining live biomass loss become CWD and cause a lagged CO_2 emission from decaying CWD

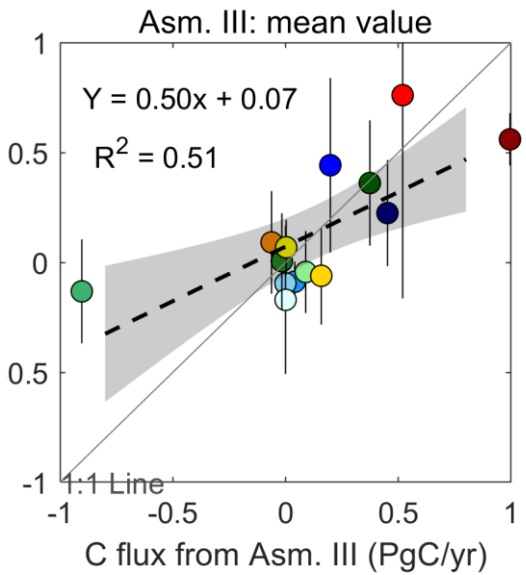


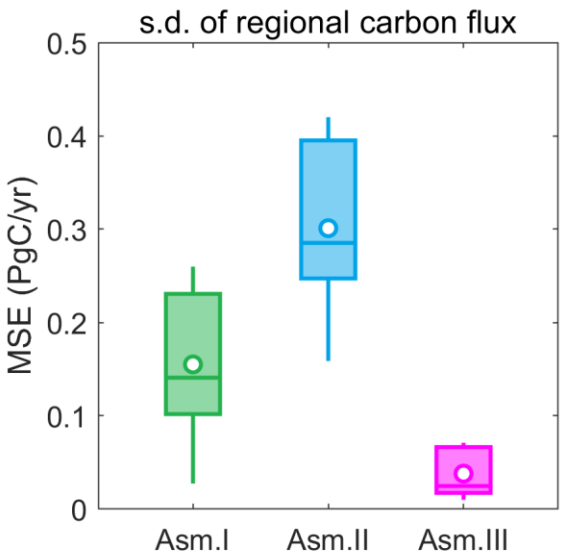
Compare with atmosphere inversion



Compare with the net biome productivity (NBP) from **nine atmospheric inversions** in the Global Carbon Project







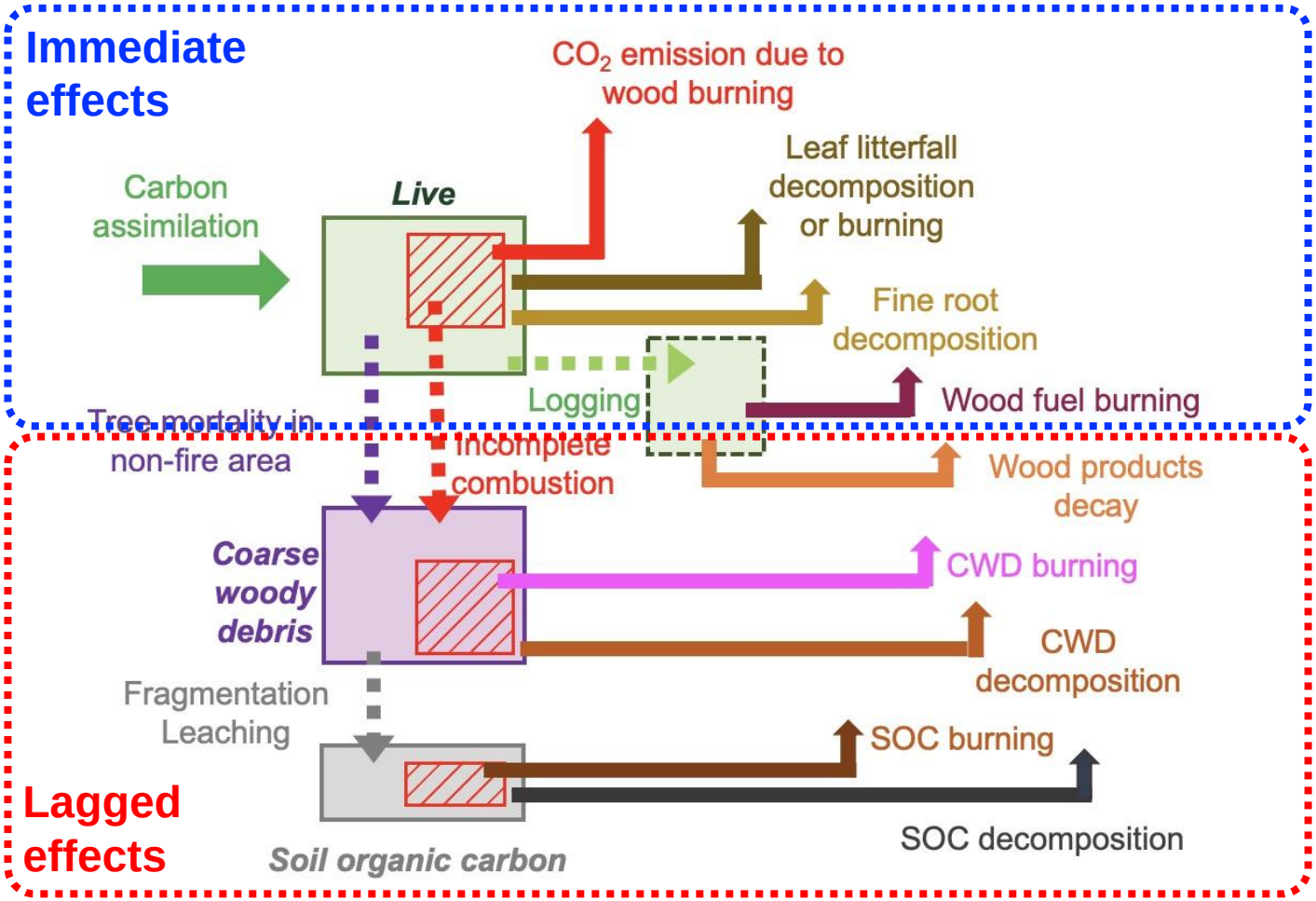
Hardly replicate the spatial variability in the mean net carbon fluxes

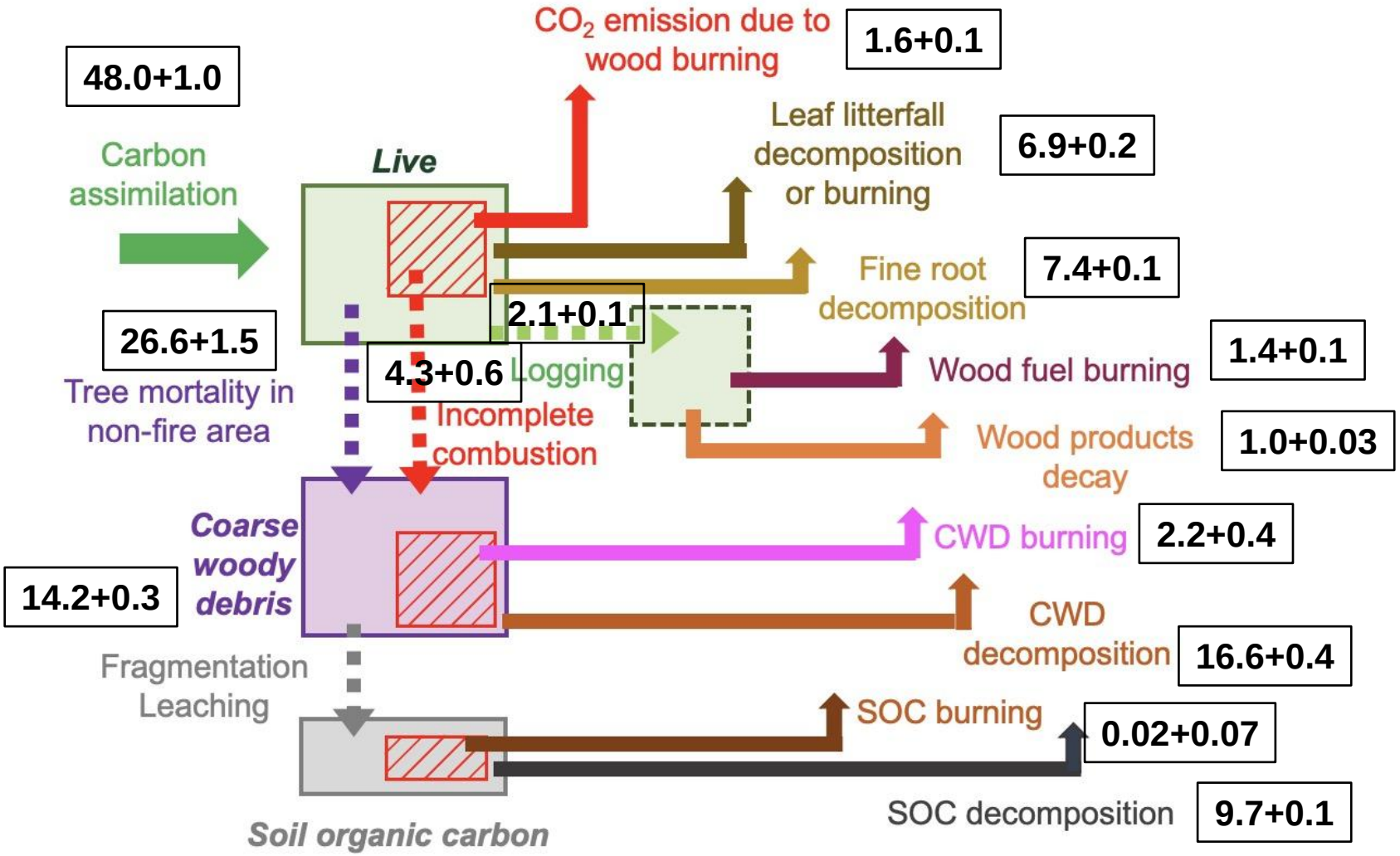
Very good performance.

Asm. III: Good agreement in the fluctuations of net carbon fluxes

Immediate effects vs. Lagged effects

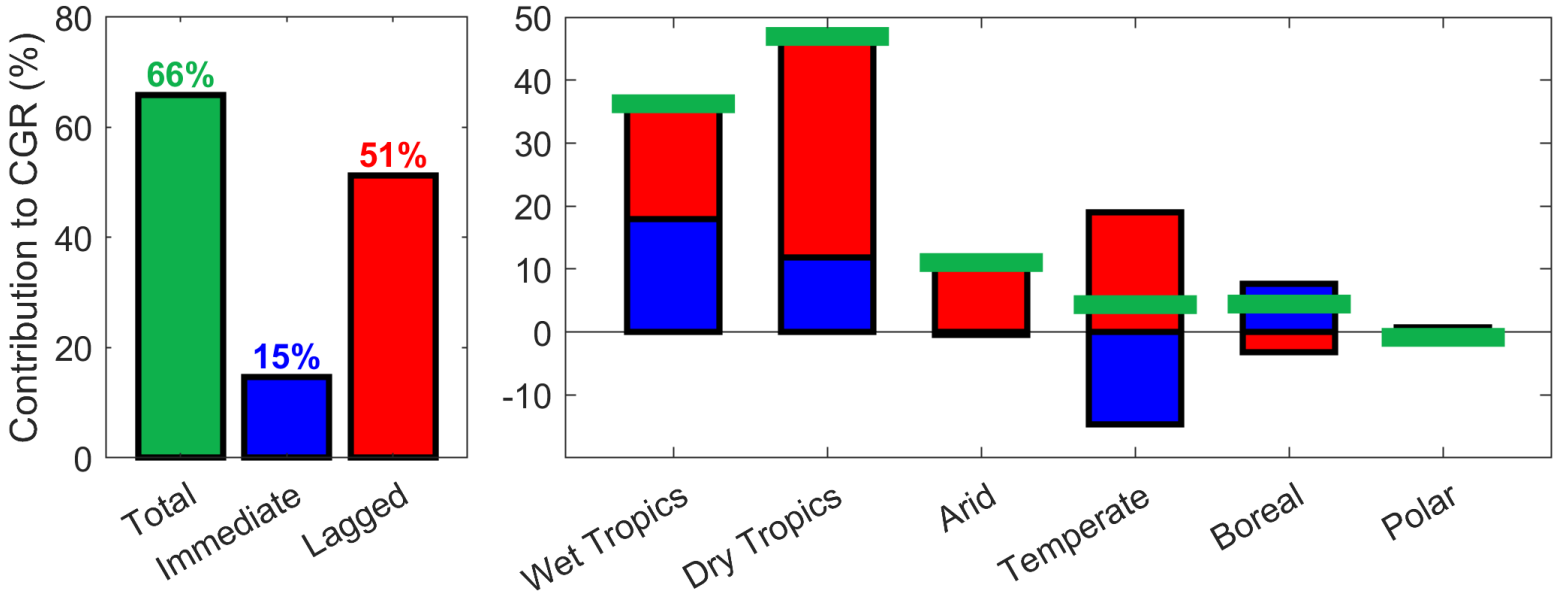
- The net carbon flux of live vegetation pool: Immediate effects
- The net carbon flux of non-living carbon pool: Lagged effects




48.0+1.0
Carbon assimilation
Live
CO_2 emission due to wood burning
1.6+0.1
Leaf litterfall decomposition or burning
6.9+0.2
Fine root decomposition
7.4+0.1
2.1+0.1
Logging
Wood fuel burning
1.4+0.1
26.6+1.5
Tree mortality in non-fire area
4.3+0.6
Incomplete combustion
Wood products decay
1.0+0.03
Coarse woody debris
14.2+0.3
CWD burning
2.2+0.4
CWD decomposition
16.6+0.4
Fragmentation Leaching
SOC burning
0.02+0.07
SOC decomposition
9.7+0.1
Soil organic carbon

Immediate effects vs. Lagged effects

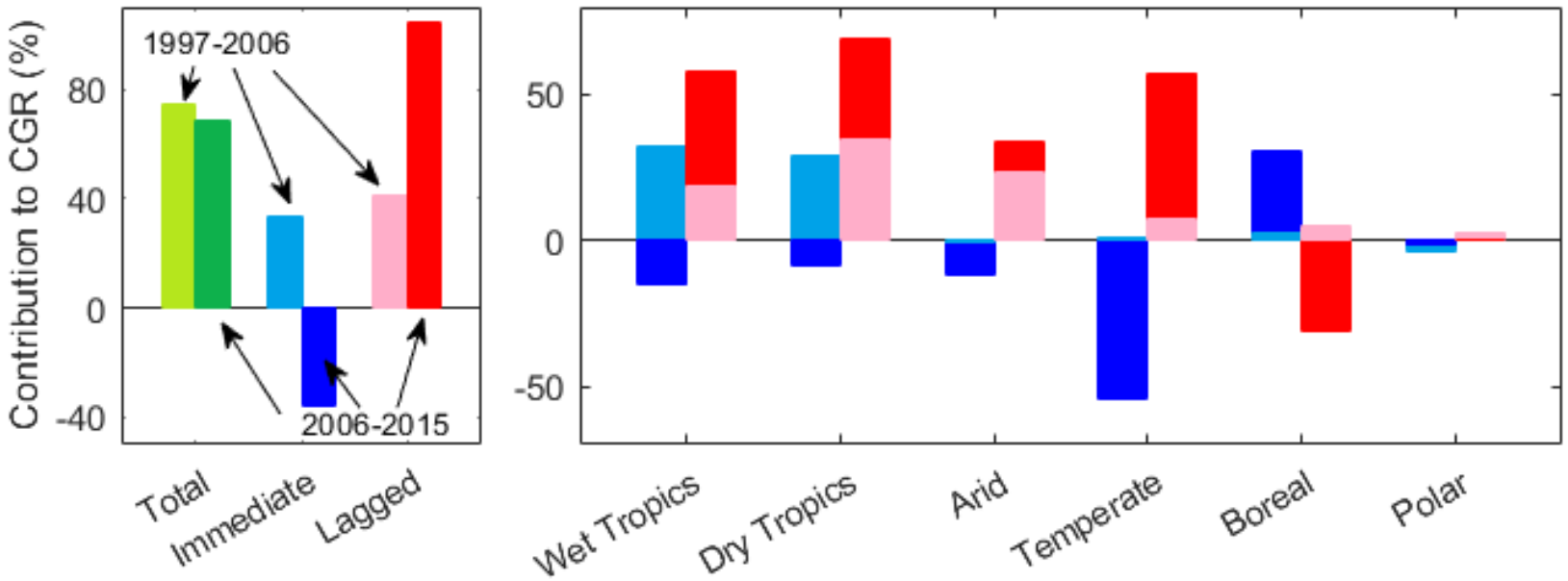
The whole period of 1997-2015



- Lagged effects contribute more than **<u>three times</u>** as much as immediate effects
- Lagged effects dominate the dry tropics while immediate effects dominate the wet tropics and boreal regions

Immediate effects vs. Lagged effects

Difference between 1997-2006 and 2006-2015



- The total contributions effects exhibited minimal changes
- The contribution of immediate effects have a substantial decrease, shifting from positive to negative
- Due to global warming, the lagged effects have doubled in magnitude

Summary

- We develop a data-driven three-pool box model to simulate carbon dynamics among live vegetation, dead vegetation and soil carbon pools.
- The modeled net carbon fluxes match well the global atmospheric CO_2 growth rate (CGR) and atmospheric inversions.
- Overall, the lagged effects of non-living carbon changes to CGR is three time the immediate effects caused by live vegetation dynamics.
- During the recent decade, the relative role of lagged carbon emissions to CGR via decomposition processes has substantially increased.